



DiabeticChain: a novel blockchain approach for patient-centric diabetic data management

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Abstract

While medical IoT data complexity surges, managing healthcare for chronic ailments like diabetes remains stagnant. Our innovative approach, DiabeticChain, not only addresses this but also introduces unique methodologies that set it apart in the realm of diabetic data management. Current healthcare architectures face significant research gaps including privacy and security concerns in sharing sensitive diabetic data, lack of patient data ownership, data fragmentation, interoperability issues, reliance on centralized authorities and ethical concerns laying the foundation for the novel solution DiabeticChain offers. Blockchain, a decentralized distributive ledger technology, can revolutionize data sharing with immutability, transparency and traceability. This paper proposes DiabeticChain, a permissioned blockchain-based, patient-centric healthcare architecture, addressing the key challenge of empowering patients to grant or revoke access to their diabetic data enabling secure sharing while safeguarding patient privacy. DiabeticChain allows data consumers to access patients' data as per their agreed-upon consent policy, using Solidity smart contracts, developed, tested and deployed using the truffle framework. A hybrid model is used for data storage to store confidential diabetic data in an off-chain secure database while keeping only metadata on the blockchain. The study includes a proof-of-concept using Go-Ethereum to establish a permissioned blockchain and Chain-Link for secure smart contract connectivity to off-chain resources. Performance was evaluated using four Azure hosted VMs and Hyperledger Caliper utilizing throughput and latency as key performance metrics to validate effectiveness. Results depict the system's feasibility with an average Query throughput of 201.7 tps, an average Transaction throughput of 200.07 tps, an average Query latency of 0.24 s and an average Transaction latency of 1.31 s. The architecture is designed to seamlessly integrate with existing EHR systems, promoting efficient data sharing within the healthcare industry.

Keywords Permissioned blockchain · Patient-centric · Smart contracts · Electronic Health Records (EHRs) · Diabetic data sharing

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1 Introduction

Diabetes is a chronic disease that requires rigorous self-care and as per the data of 2021, there are 537 million adults living with the disease globally and this number is predicted to rise at a pace higher than expected resulting in 643 million by 2030 and 783 million by 2045 [1]. Despite the ubiquity of diabetes as a global health concern, recent advancements in data sharing remain inadequate. Our research identifies these gaps, laying the foundation for the novel solutions DiabeticChain offers. Even though people with diabetes have lower life expectancy than the ones without the disease, the life-time medical cost of diabetes is still significant [2]. Patients diagnosed with this disease need to take daily care and are responsible for the choices of diet, exercise, regular medication and testing of blood sugar levels. The current state of diabetes care is worsening due to challenges like aggravated patient-to-provider ratio, the presence of geographical barriers for diagnosis and increasing patient and provider burden due to absence of effective diabetic data sharing [3]. The self-management of diabetes plays a crucial role in complex patient care and continuous monitoring of health data is important [4]. Along with regular medication and diabetes diet plan, patients have to monitor their blood glucose levels, weight and blood pressure regularly. Huge amounts of data are generated and utilized during this process, and this measured data has to be effectively shared with medical practitioners to ensure patient care. Thus, data sharing plays a crucial role in managing the disease [5].

Whenever patients visit a medical practitioner, their data is gathered and stored as EHRs in centralized databases, which are owned, managed and controlled by central authorities like hospitals. Electronic Health Records (EHRs) are patient records stored digitally consisting of health information like medical history, current medication, allergies, test reports, etc. maintained for better patient diagnosis and treatment by healthcare professionals [6]. Even though we have shifted from document-based health charts to EHRs, the data is still stored on central servers [7]. These central authorities may share the patient data of interest with various stakeholders for better patient care or advancing research. Patients have no control over their private and sensitive data raising serious concerns for security and privacy. According to the Health Insurance Portability and Accountability Act (HIPAA) [8], every patient has the right to own their health information and set rules to give consent to share their health data with other entities. This leads to the requirement of a decentralized system where patients are the complete owner of their data, thereby having control over who can access their data.

Due to the complexity and privacy of the EHR data, it should be restricted from unauthorized access. With the increasing patient mobility and the development of multiple healthcare facility providers, it has become easier for patients to consult multiple hospitals and doctors for the treatment of chronic diseases like diabetes [9]. The data is thus becoming fragmented due to being stored in multiple databases, making it difficult for doctors to access the complete medical history of the patient. Traditionally, if one hospital requires to share the patient's data with another hospital, then the patient signs a consent form stating the

detailed use of data shared and this tedious process is carried out manually, making it time-consuming [9].

Many challenges exist that hinder the traditional way of sharing diabetic data and need to be addressed. These are security and privacy concerns in sharing sensitive diabetic data, no data ownership for the patient, data fragmentation, lack of interoperability, requirement of trust in a centralized authority and ethical issues. DiabeticChain presents a novel approach to address these fundamental issues in diabetic care. The main challenge addressed in this paper is the protection of privacy of the patient data by giving them control to grant or revoke access to their data enabling secure sharing of diabetic data among stakeholders. Unlike the conventional models, DiabeticChain prioritizes patient data ownership, data privacy and secure data sharing.

1.1 Background and motivation

Healthcare data is today captured and generated from multiple sources ranging from the patients themselves to wearable devices like smartwatches, smart bands, smartphones, etc. With increasing technological advancements, IoT devices like wireless sensors are being used to monitor continuous patient health, generating huge amounts of personal, valuable and complex data. If controlled access to complete data is provided, then it can substantially benefit the entire healthcare system. Since patient data is owned by trustless third-party authorities like hospitals, laboratories and health insurance companies, secure and efficient data sharing is not possible. There exists no method for verifying the medical data sharing by hospitals to an untrusted third party [10]. This creates vulnerabilities in data security, increasing the possibility of unauthorized access.

There also exists a possible risk of a data breach due to the existence of a single point of failure. According to existing reports, 89% of users using IoT in healthcare have suffered security breaches and 80% of them are still willing to use this technology [11]. Unauthorized access through malicious attacks on diabetic healthcare data can have disastrous results like medical identity theft, insurance fraud and tampering with ongoing treatment or medical history. A secure and efficient digital EHR data sharing system is thus required. To overcome the limitations of conventional data sharing models discussed above, a decentralized approach to medical data sharing can potentially provide patients with ownership over their data, thereby providing more control over their data, helping in eliminating a single point of failure in centralized systems, minimizing cyberattacks and building trust among the stakeholders involved in diabetic care.

There are some unique challenges as well associated with diabetic data like diabetes is a disease that requires self-care by substantially burdening the patients with continuous glucose monitoring and carbohydrate counting [3]. As per the research in [12], an average adult with type 2 diabetes is recommended to spend 4 h per day as the total estimated time for self-care. Being a chronic disease, diabetes also burdens the healthcare providers to analyse and access the huge amount of data associated with it [3]. While many researchers have worked in the field of medical data

sharing, there exists a need to specifically address these unique challenges associated with sharing of diabetic data among stakeholders. This gap in the existing literature serves as the primary motivation of this research.

1.2 Objectives and contributions

Blockchain is a decentralized distributive ledger technology that has the potential to revolutionize the way confidential data is shared and accessed today by providing benefits like immutability, transparency, traceability and enhanced security [13]. In this paper, we propose a system—DiabeticChain, a patient-centric healthcare architecture to support secure and efficient data sharing of diabetic data, provide an implementation as a proof-of-concept study and then verify its performance using various assessment metrics. The application of a one-time access URL to access patient data according to their consent, makes our approach novel and a necessary solution.

The aim is to address the current limitations of the traditional data sharing models by using a decentralized technology based on blockchain for diabetic data sharing. DiabeticChain's goal is to solve the requirements of patient privacy, data security, user control over data, traceability, data integrity, interoperability and standardization. The following are the major contributions of this paper:

- Proposed a novel blockchain-based patient-centric healthcare architecture for safe storage and secure diabetic data sharing among stakeholders like healthcare providers, laboratory technicians, health insurance organizations and researchers.
- Implementation of an architecture where data consumers can access patient's diabetic data as per their agreed-upon access policy using a one-time access URL implemented by self-executing smart contracts written in Solidity to ensure the controlled exchange of sensitive data based on the blockchain.
- Evaluated the performance of the proposed architecture based on throughput and latency assessment metrics to ensure its feasibility and reliability in real-world health scenarios.

While many blockchain-based solutions mainly focus on the data sharing between the healthcare entities, DiabeticChain places patients at the centre of the healthcare architectures by empowering them with the ability to control who has access to their data. The one-time access URL system ensures that the patients explicitly grant access as per the predefined agreed-upon access policy. This level of consent sharing makes it different from the traditional healthcare data sharing models.

1.3 Organization

The remaining paper is organized as follows: Sect. 2 summarizes and compares the related work. Section 3 provides an overview of the technical background of blockchain. Section 4 describes the blockchain potential in diabetic data sharing.

Section 5 states the design requirements and the proposed architecture is discussed in Sect. 6. Section 7 discusses the implementation of the workflow. Results and performance evaluation are described in Sect. 8. Section 9 concludes the paper and discusses the limitations and scope for future work.

2 Related work

Promising research has been carried out in the field of blockchain-based healthcare solutions to support healthcare data sharing to further address the concerns of privacy and security, data fragmentation, lack of interoperability, lack of trust in healthcare data sharing and no ownership of data in the hands of the patients. Let us examine the challenges addressed by different researchers and how their solution is different.

2.1 Blockchain-based healthcare solutions to support healthcare data sharing

In [14], Zyskind and Nathan employed blockchain, re-purposed as an access-control moderator and an off-chain data storage solution to ensure that users control and manage their data protecting users' privacy. The authors in [15] proposed a decentralized blockchain-based framework MedRec using the Ethereum blockchain to share and manage Electronic Medical Records (EMRs). It can manage confidential information and allows authentication, accountability and secure sharing of medical data. But the framework uses private databases to store data which is prone to cyber-attacks and will also allow an internal entity to have access to the sensitive data.

The authors in [16] proposed a patient-centric permissioned blockchain-based medical data sharing to ensure privacy and patient identity management using Hyperledger Fabric. But the authors did not address the user authentication to secure data storage. In [17], the authors designed a system to share and manage personal health data using blockchain and cloud storage, eliminating the dependence on a centralized third party for sensitive data sharing. However, the proposed system is yet to be integrated with the existing healthcare systems.

Dubovitskaya et al. [9] designed a framework named ACTION-HER using a permissioned blockchain and Hyperledger Fabric to allow patients to control the sharing of their health records securely for radiation oncology. This approach enabled healthcare providers and researchers to access patient data more quickly and efficiently while significantly reducing costs. However, the limitation of this paper was the risk of having a single point of failure of the system and in case of emergency situations, there is a requirement of "break glass" mechanism.

Jabarulla et al. [18] implemented a distributed patient-centric image management system using blockchain to ensure security, safety and privacy of patient data without using centralized authority. Due to manual encryption and decryption of original medical images the system poses a risk of losing private keys. Mani et al. [19] proposed a patient-centric approach to store health records on an IPFS-based distributed ledger system using Hyperledger Fabric technology and resolved privacy,

security and confidentiality concerns to enable decentralized storing and sharing of health information while ensuring scalability. A large number of resources are required for implementation of multi-blockchain systems and extended frameworks integrated with Non-Fungible Tokens.

The authors in [20] proposed a novel secure and efficient patient-centric healthcare framework (PCH) reference architecture that leverages blockchain, cloud and IoT technologies to improve semantic interoperability in healthcare systems. However, the costs associated with the maintenance and implementation of the system could be high and do not address the legal and regulatory challenges associated with sharing patient data across different healthcare systems. Taylor et al. [21] proposed VigilRx, a patient-centric and interoperable prescription management system that ensures user control, prevents information blocking and improves transfer efficiency ensuring the security of patient data, interoperability, fraud prevention, data accuracy and accessibility. The system is heavily dependent on blockchain which limits its effectiveness by scalability and security of the network making it vulnerable to blockchain congestion and security breaches.

In [22], the authors proposed a patient-centric blockchain-based health record access framework using Hyperledger Fabric to enhance patient control, protect privacy and data integrity. Patients can impose necessary conditions before granting access to their medical data. But they proposed a conceptual framework that has not been implemented in a real-world setting and integrated with the existing healthcare models. Rai [23] proposed a patient-controlled blockchain-based Electronic Health Records management system to ensure patient ownership of their medical data, elimination of security risks, data integrity and confidentiality. But the smart contracts of the framework are based on time and notifications.

The existing solutions to support healthcare data sharing face limitations like scalability, integration with the existing healthcare systems, high cost of implementation and requirements of every healthcare facility on the blockchain network. Table 1 summarizes the existing literature. In comparison with the existing works, we have proposed a feasible architecture for diabetic data sharing among stakeholders where data is shared as per the patient's access policy, ensuring privacy and security of data using a one-time access URL for data access and implementing the same using Ethereum blockchain and smart contracts written in Solidity.

2.2 Existing patient-centric healthcare systems to support diabetic data sharing

Traditionally, a client/server approach was followed for sharing healthcare data where a centralized server like hospital database stored the entire patient data and was solely responsible for controlling and managing it. In this approach, patients have no control over their data and have to trust the centralized authorities for ownership of their data. Table 2 depicts the differences between traditional and blockchain-based data sharing models.

In [24, 25], the authors proposed a web-based portal to link the patients' Electronic Health Records (EHRs) stored on the centralized hospital server to provide easy access. But such approaches raise concerns about privacy and

Table 1 Summary of existing literature

Authors	Problem statement	Challenges addressed/advantages	Limitations
Zyskind et al. [14]	Blockchain-based decentralized personal data management system	Data ownership, transparency and auditability of data, access control	Performance evaluation has not been done
Azarria et al. [15]	Blockchain-based framework MedRec, to share and manage EMRs	Authentication, accountability, secure sharing, interoperability of data	Data is stored in private database
Liang et al. [16]	User-centric health data sharing solution to privacy issues	Privacy, access control, data integrity, scalability	User authentication to secure data storage not considered
Chen et al. [17]	Secure management of medical data using blockchain and cloud storage	Privacy protection, secure storage, no dependence on centralized power	Not yet integrated with existing healthcare systems
Dubovitskaya et al. [9]	Patient-centric blockchain-based management of EHR for radiation oncology data	Patient centrality, secure data sharing	Single point of failure risk, “break-glass” mechanism required for emergency situations
Jabarulla et al. [18]	Implemented a distributed patient-centric image management system using blockchain	Security, safety and privacy of patient data without using centralized authority	Manual encryption and decryption of original medical images, risk of losing private keys
Mami et al. [19]	Patient-centric approach to store health records on an IPFS-based distributed ledger system using Hyperledger Fabric technology	Resolved privacy, security, confidentiality concerns to enable decentralized storing and sharing of health information while ensuring scalability	Large number of resources are required for the implementation of multi-blockchain systems and extended frameworks integrated with Non-Fungible Tokens
Gohar et al. [20]	Patient-centric healthcare framework (PCH) using blockchain, IoT, and cloud technologies to improve interoperability	Lack of semantic interoperability, patient centrality	Cost of implementation and maintenance could be high
Taylor et al. [21]	Proposed VigilRx, a patient-centric and interoperable prescription management system that ensures user control, prevents information blocking and improves transfer efficiency	Ensuring the security of patient data, interoperability, fraud prevention, data accuracy and accessibility	Heavily dependent on blockchain limiting its effectiveness by scalability and security of the network, vulnerable to blockchain congestion and security breaches

Table 2 (continued)

Authors	Problem statement	Challenges addressed/advantages	Limitations
Abutaleb et al. [22]	Integrity and privacy aware, patient-centric health record access control	Lack of control, transparency, data integrity, privacy	Conceptual framework, not been implemented in a real-world setting
Rai [23]	Proposed patient controlled blockchain-based Electronic Health Records	Patient ownership of their medical data, elimination of security risks, user authentication, data integrity and confidentiality	Smart contracts based on time and notifications

Table 2 Difference between traditional and blockchain-based data sharing models

Basis	Traditional data sharing models	Blockchain-based solutions
Patient control	Data is stored on a centralized server; patients have no control over it.	Being a decentralized network, patients have complete control over their data.
Security	Centralized servers are prone to malicious attacks.	Data encryption and consensus protocols ensure security.
Privacy	Patients cannot track who is using their data.	Traceability and user authentication ensure privacy.

confidentiality of healthcare data due to the existence of a single point of failure. For optimization of medical data storage and easy real-time sharing of data, the approach was shifted from web-based to cloud-based healthcare records management where data is stored in cloud attribute-based encryption (ABE) servers. As discussed in [26, 27], the limitations of this approach include the possibility of cyberattacks making them vulnerable to privacy, the need for large storage capacity, high computing power so that millions of patients can access the server at once and the data is still not owned by the patients.

To address the challenge of data ownership by patients, patient-centric blockchain-based solutions are introduced to support secure and efficient diabetic data sharing and are a need of the hour. In [28], the authors proposed a patient-centric architecture for management of medical data using blockchain where patients can track their data and how it is shared among stakeholders. In [29], the authors proposed a feasible solution to shift the data ownership from healthcare providers to the patients by incorporating the unique features of blockchain technology, but this solution required a setup at every healthcare facility and faced the issue of scalability. In [5], the authors proposed a platform to capture diabetic health data using IoT devices and share it with health entities using blockchain to protect patient ownership and privacy of data, but the paper has not been implemented and integrated with the existing healthcare system.

Previous work on diabetic data management has predominantly focused on specific aspects such as pandemic response [30], prediction models [31] and remote monitoring [32]. Subramanian et al. [30] emphasize on pandemic response and consortium-based management of diabetes-related healthcare to efficiently prioritize the needs of diabetes patients during a pandemic. Hennebelle et al. [31] focused around diabetes prediction using IoT, AI and blockchain for data security based on risk factors. Pradhan et al. [32] focus on remote patient monitoring, security, the use of smart contracts and the automatic collection of sensor data. In contrast, our paper, DiabeticChain, offers a comprehensive solution addressing patient's diabetic data privacy, control and integration with existing Electronic Health Records (EHRs), differentiating it from existing approaches.

2.3 Challenges and limitations

The traditional models implemented for sharing health records and their management are raising concerns about security and privacy of healthcare data due to the presence of trustless centralized servers. Data is owned by centralized authorities like hospitals. Patients are not aware of how their data is being used and they cannot track who is using their data, becoming vulnerable to internal and external attacks. These challenges faced by the existing healthcare data sharing models need to be addressed by leveraging the benefits of blockchain like transparency, immutability and decentralization. The proposed DiabeticChain system addresses these challenges by using a permissioned blockchain, patient's consent policy and a one-time access URL to share diabetic data. Table 3 summarizes the literature on

Table 3 Comparison of existing literature

Study	Existing challenges					Use case	Type of blockchain	Technology	Data storage	Performance evaluation
	I ^a	II ^b	III ^c	IV ^d	V ^e					
Azaria et al. [15]	✓	✓	✓	✓	✓	Electronic Medical Records	Permissioned	Ethereum	Off-chain	×
Chen et al. [17]	✓	✓	✓	✓	×	Healthcare Data Sharing	Permissioned	–	–	×
Dubovitskaya et al. [9]	✓	✓	✓	✓	✓	Radiation Oncology Data	Permissioned	Hyperledger Fabric	Mixed	×
Jabarulla et al. [18]	✓	✓	✓	✓	×	Medical Image Management	Permissioned	Ethereum	On-Chain	✓
Mani et al. [19]	✓	✓	✓	✓	✓	Healthcare Data Sharing	Permissioned	Hyperledger Fabric	On-chain	✓
Gohar et al. [20]	✓	✓	✓	✓	✓	Healthcare Data Sharing	Permissioned	Hyperledger Fabric	Mixed	×
Taylor et al. [21]	×	✓	✓	✓	✓	Prescription Management System	Permissionless	Ethereum	On-chain	✓
Abutaleb et al. [22]	✓	✓	✓	×	✓	Healthcare Data Sharing	Permissioned	Hyperledger Fabric	–	✓
Rai [23]	✓	✓	✓	✓	✓	Electronic Health Records	Permissionless	Ethereum	On-Chain	✓
This study	✓	✓	✓	✓	✓	Diabetic Data	Permissioned	Ethereum	Mixed	✓

a: Data Security, *b:* Patient Privacy, *c:* User Control over Data, *d:* Traceability, *e:* Interoperability

the current healthcare data sharing architectures and how this paper is different from the existing work.

While there is no dearth of blockchain-based architectures in healthcare, most fall short in X, Y and Z aspects. DiabeticChain, in contrast, excels in these areas, offering a more comprehensive and innovative solution. These proposed architectures may not model a feasible design that can be combined with the existing Electronic Health Records (EHRs) approach in healthcare systems for secure and efficient data sharing and promoting interoperability. In order to integrate it with existing healthcare systems, the major challenge is lack of interoperability which allows the decentralized servers to communicate securely over a blockchain network and data format standardization. The data captured from EHR needs to be securely transferred without tampering and access should be controlled. There may exist certain technological barriers among other challenges mentioned above to completely shifting the traditional EHR data to blockchain and fully implementing blockchain-based solutions, but it does not seem much far-fetched today.

Being a new technology, blockchain-based solutions still face certain issues like the obstacle of scalability and implementation challenges like technical, organizational and regulatory concerns, but there is a significant growth in the number of proposed solutions to address these existing limitations [33]. There exist potential limitations specific to the implementation and adoption of the DiabeticChain system in the existing healthcare system such as scalability, data standardization and regulatory issues. As the size of the network increases and more users participate as nodes, higher computational power is needed for implementation. The system may also pose regulatory issues as there does not exist any central authority that can be held accountable [34].

3 Technical background

This section gives an overview of the technical background of Blockchain technology describing its architecture, the different types of blockchains used in applications and how blockchain supports dynamic consent which forms the backbone of this paper.

3.1 Overview

Blockchain is an immutable, decentralized and shared ledger that facilitates secure storage of transactions and tracking of assets by enabling a network of authorized nodes and eliminating the requirement of trust in any third party or among the nodes [35, 36]. It has a backbone in cryptocurrency and was first introduced in financial sector as a peer-to-peer cash system with the implementation of Bitcoin by Satoshi Nakamoto [37]. Since then, it has been incorporated by many other sectors for numerous applications and has the potential to revolutionize different sectors like the financial institutions, education, voting systems, healthcare industry and many other trustless environments. Overall, the healthcare industry can be revolutionized using

blockchain to enhance patient care, reduce costs of redundant diagnosis, ensuring privacy and security of healthcare data and its efficient sharing among stakeholders [38].

In the simplest terms, blockchain consists of blocks chained together to form a distributed decentralized database. Every block stores a timestamp and a hash of the previous block computed cryptographically, making it tamper-proof as even a small modification in any block will modify the hash value of the subsequent blocks and thereby affecting the entire blockchain and the alteration can be easily detected. A single block's structure in the blockchain network is depicted in Fig. 1. There is a header and a body for every block. All of the data nodes are hashed in a Merkle tree found in the body. A timestamp, nonce, Merkle root, hash pointer to the previous block, transaction counter and version information are all included in the header, which serves as the framework for the blockchain.

Blockchain technology is ideal to use where trust is required without the presence of a third party because any new transaction or changes to the data require consensus among all the participants of the nodes. The cooperation among nodes to validate a transaction is established using a consensus protocol and thereby trust is upheld. The most widely used consensus algorithms are Proof of Work, Practical Byzantine Fault Tolerance, Proof of Stake, etc. In Proof of Work (PoW), the validators must compete to solve a mathematical puzzle or a cryptographic equation to generate a new block and get rewards, while in Proof of Stake (PoS), the rights within the network depend on the time for which a node holds the coins and the number of coins held.

Blockchain works by implementing the self-executing smart contracts which consist of rules of an agreement between a seller and a buyer written into lines

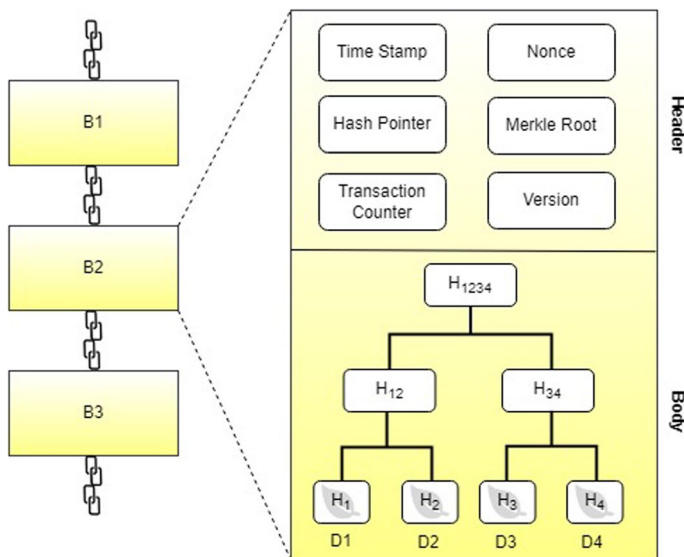


Fig. 1 Structure of a block

of code. The core idea behind them is to make the process of security breach so expensive that it results in secure agreements over public networks without trusting a third party [39]. After the execution of a smart contract, a transaction is appended on the blockchain.

3.2 Blockchain architecture

The architecture of a blockchain consists of two main components—a distributed ledger and peer-to-peer network. Figure 2 illustrates the architecture of a blockchain as a peer-to-peer network and how it differs from the traditional centralized network.

- **Distributed Ledger**—All the transactions taking place in the blockchain network are recorded in a ledger shared by all participating nodes eliminating the possibility of some malicious node intervening and manipulating the records. Every node has an identical local copy of that shared ledger to ensure the data integrity. To append a new transaction to a block in the blockchain, it is first verified and authenticated by all nodes using a consensus algorithm and then appended to the network only if majority of the nodes validate the transaction. Once this change is made it cannot be reverted. No single node has the authority to manipulate the data existing in the blockchain.
- **Peer-to-peer network**—In a blockchain network, the nodes are connected instead of one central node, making the entire network decentralized and every transaction is authorized before adding it to the network, making the network authentic [40]. Every node has the equal power to consume the services of the network and

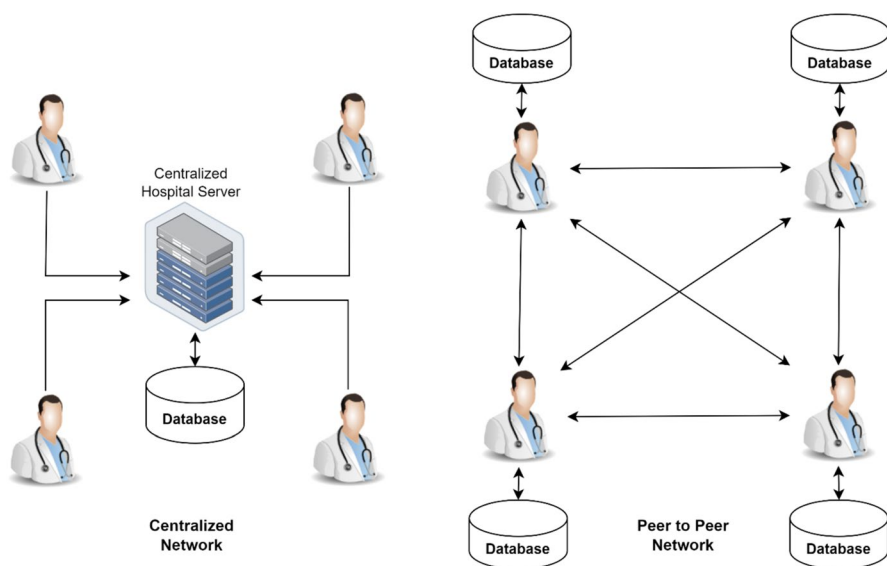


Fig. 2 Blockchain architecture

agree upon them via a consensus algorithm. Peer-to-peer networks eliminate the problem of a single point of failure due to absence of a central node, resulting in a more secure network [41].

3.3 Types of blockchains

There are different ways to establish a blockchain network depending on the applicable use case and their requirements such as cryptocurrencies like bitcoin have a public network where any node can participate and verify the transactions, but on the other hand a bank would require a private network where only the authorized users can participate and access the confidential information [42]. To meet such varied needs, blockchain is categorized as private and public. In public blockchains, there is no restriction and anyone can access the network to validate the transactions. No single entity has the complete control over the network and power is equally divided among nodes. While in private blockchains, certain nodes are authorized by the network administrator to participate in the network and access is restricted to them. Blockchain is also categorized as permissioned and permissionless to differentiate between who can generate a new block in the blockchain network. In a permissioned blockchain, among the participating nodes, only some are granted the permission to participate in the consensus algorithm to write a new block. In case of a permissionless blockchain, any node of the blockchain network can participate as a validator [43].

Blockchain can thus be divided into 4 different types as per the role of participating nodes in the network [44, 45]. Each type is described in detail as follows:

1. **Private permissionless:** Only a restricted number of authorized nodes can enter the network and all participating nodes can perform the read or write operations in the network to create a new block. Equal importance is given to every node. An example of private permissionless blockchain is Holochain [46].
2. **Private permissioned:** Only a restricted number of authorized nodes can enter the network and all participating nodes can read the contents of the blockchain, but a subset of them is given the permission to perform the write operation to generate a new block in the blockchain. Some examples of the private permissioned network are Hyperledger Fabric [47] and Hyperledger Besu [48].
3. **Public permissionless:** Any node can enter the blockchain and perform read or write operations in the network. In this case, in order to build trust among the participants, costly consensus protocols are required. Some examples of public permissionless blockchains are Ethereum, Bitcoin and Witnet [49].
4. **Public permissioned:** Any node can enter the blockchain and read the contents present on the network, but only certain authorized users/nodes are allowed to perform the write operation. They need less costly consensus protocols to build trust among participants. Some examples of public permissioned blockchains are ESIO [50] and Ripple protocol consensus algorithm [51].

Table 4 summarizes the different types of blockchain.

Table 4 Types of blockchains

Types of blockchain	Private	Public
Permissionless	Restricted participating nodes, All can read and write. For example, Holochain	Access to all, anyone can read and write. For example, Ethereum
Permissioned	Restricted participating nodes, All can read, a subset of them can write. For example, HLF	Anyone can read, a subset of them can write. For example, Ripple

3.4 Dynamic consent and blockchain

Dynamic Consent is the approach that involves collecting, storing and managing of data obtained from people for the purpose of research and healthcare. This approach is very different from the traditional consent in which the data was collected from the people only at the beginning of a study that is taking place but on the other hand dynamic consent approach involves dynamic or continuous flow or control of the data for various research activities or purposes. Despite such flow, dynamic consent has limited dimensionality in the practical field due to the security and legality of the personal data belonging to people. This poses major trust issues for the people including the fear of maintaining the confidentiality [52, 53] of the data that belongs to their personal belongings. Now, blockchain deals with the rapidly multiplying part of data in the form of a decentralized database [54]. Blockchain has various functionalities which could support dynamic consent in different ways like high accountability, a greater sense of data security and privacy, immutability and an extremely efficient management system [22]

Diabetic data sharing takes place in a continuously changing environment, and therefore, dynamic consent is necessary for personal protection of sensitive data, preventing the unforeseen risks of privacy and driving patient-centric research [55]. Blockchain can enable data sharing in a trusted environment according to the modifiable consent preferences of the patient where they can dynamically update their consent options anytime.

4 Methods

This section discusses the requirement of blockchain to provide a viable solution for the secure and efficient sharing of diabetic data among stakeholders and the formatted research questions for this study.

4.1 Blockchain potential in diabetic data sharing

There are various requirements for the construction of a functional system, and in the scenarios of handling such vast and categorical data on diabetes, we need to consider optimal methods. Sometimes these methods depend on situational scenarios. And, determining whether any particular method is suitable for a particular application is an uphill battle. Therefore, we can say that deciding whether blockchain will be able to simplify the scenario is not correct [22]. To handle this, Wüst and Gervais [56] proposed a decision tree and Pedersen et al. [57] proposed a decision path in order to discover the applicability of blockchain. This method made it concise to 6 questions that need to be answered for diabetic data sharing condition in order to determine the need for blockchain as depicted in Fig. 3.

1. Does the state need to be stored? The answer to this question is dependent on the consideration of the following factors:
 - Patient Consent: In diabetic data sharing system, it is inevitable to store the consent of the patient so as to allow secure access to their data from interested data consumers. For instance, if any researcher wants access to patient diabetic data, then the consent policy is validated before granting the access.
 - If a chronic disease is to be diagnosed: For instance, if any person has been diagnosed with any disease which is chronic like diabetes and is previously diagnosed with another implication of the same category of disease then, there is a need to store the state for further approximation of such disease since it requires ongoing medical attention. And blockchain could inculcate this very well.

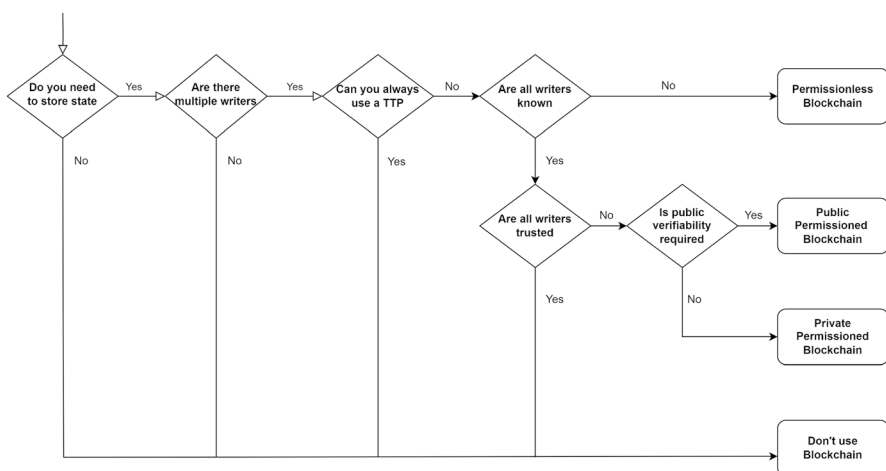


Fig. 3 Decision path of using blockchain

- The number of times the data needs to be handled: For instance, if there is a single-time utility of the data of any patient who has been diagnosed with diabetes, then there is no need to store the state, while, if there is a need to create a system that diagnoses and remembers the patient's personalized way of attending to data, then there is a need to store the state. In our case, diabetes is a chronic disease, and hence, the diabetic data is handled multiple times.
2. Are there multiple parties involved? In general, this question can be answered in a Yes. Since there are many contributors towards the health-related information of patients. For instance, patients themselves could contribute to their physical measures like height, weight, blood pressure levels, blood oxygen levels, etc. And laboratory technicians could contribute to their in-depth blood test results diagnosis and different scans of their body, while doctors and researchers could add their actual physical condition and the diseases or problems that patients have in their bodies. Hence, using blockchain all the data inputs by various parties could be maintained and handled well.
 3. Can you use a centralized trusted third party? The answer to this question is again partially, both Yes and No and the reasons behind them are as follows: Yes, because it is possible to always use a web-based trusted third party for diabetic data sharing. As a third party might act as a platform that could inculcate multiple users and make the process even smoother and make the data sharing relatively easier by adding a small functionality of secure login ID and Passwords. Another advantage of being a consolidated platform is that it could prevent unusual data redundancy and corruption. While the answer is no, in the terms of the betrayal of the patients by sharing their personal details with any web-based third party. Patients cannot trust the third party every time and track the use of their diabetic data and blockchain is able to maintain the trust without using the third party.
 4. Are all writers known? Yes, all writers like patients, data controllers and data consumers should register in the system before interacting. In a diabetic data-sharing system, it is necessary to have known writers like patients, healthcare professionals such as laboratory technicians, etc., researchers and doctors who are legally approved to access the data of the patients. All these writers must have their authenticated login credentials, and also, they must be posed to access the data of their radii (data of their utility).
 5. Are all writers trusted? The answer to this will be in No, all the writers cannot be trusted. However, all the patients have a mutual sense of trust in all the healthcare professionals. But these writers must comply with their boundaries and must be regularly checked for their utility with their data. This could be ensured by blockchain by using encryption models and security protocols.
 6. Is public verifiability required? Public verifiability means the involvement of external auditors to hamper the critical data of patients. In spite of being a desirable feature, public verifiability is strictly not required and blockchain is playing its part by keeping the data-sharing system private.

4.2 Research questions

The following research questions were developed after analysing the existing blockchain-based patient-centric healthcare data sharing models and need to be addressed with respect to the diabetic data sharing use case.

1. What are the limitations and challenges of the existing healthcare data sharing models for diabetic patients? There are unique challenges associated with diabetic data since diabetes is a disease that requires self-care by burdening the patients with continuous glucose monitoring and the healthcare providers to analyse and access the huge amount of data associated with it [3]. The current state of diabetes care is worsening due to absence of effective diabetic data sharing. The traditional systems face privacy and security concerns in sharing sensitive diabetic data, no data ownership for the patient, data fragmentation, lack of interoperability, requirement of trust in a centralized authority and ethical issues.
2. How can blockchain-based solutions address the limitations and challenges of the existing healthcare data sharing models for diabetic patients? The challenges faced by the existing healthcare data sharing models can be addressed by leveraging the benefits of blockchain like transparency, immutability and decentralization. The main challenge addressed by DiabeticChain is the protection of privacy of the patient data by giving them control to grant or revoke access to their data enabling secure sharing of diabetic data among stakeholders by using a permissioned blockchain, patient's consent policy and a one-time access URL to share diabetic data.
3. What performance metrics can be used to evaluate the effectiveness of a blockchain-based solution for diabetic data sharing? The performance of blockchain-based solutions for diabetic data sharing can be measured in terms of the transaction and query operations using throughput and latency as performance metrics to validate their effectiveness. Measuring factors such as the number of transactions per second, the response time for transaction processing and throughput can be used to evaluate the system's feasibility.

5 Design requirements

In order to implement our proposed architecture, five design requirements have been identified in accordance with our blockchain use case.

5.1 Requirement 1: data security

Data security is a crucial requirement in sharing sensitive diabetic data which is susceptible to cyberattacks. Both active (taking control of medical devices) and passive (overhearing the communication through wireless devices) attacks can occur on the collected diabetic data [58]. Patients' data should be stored in a secure

database to eliminate the possibility of malicious attacks and prevent the misuse of data by attackers. Data encryption techniques like public key cryptography and access control mechanisms have to be employed to eliminate the unauthorized access and hence ensure data security [9, 15, 16, 59, 60].

5.2 Requirement 2: patient privacy

Patients' diabetic data should be shared with healthcare professionals and researchers with proper authentication while minimizing the disclosure of their identity and without compromising their privacy. Only the authorized users should have access to the confidential diabetic data of patients with their consent [16, 22, 61, 62].

5.3 Requirement 3: user control over data

Healthcare has moved from document-based storage to EHRs, but the ownership and control of confidential data is still in the possession of unknown, untrusted, centralized authorities like hospitals with a possible risk of data breaches [7]. Diabetic data of the patients should be owned, controlled and managed by the patients themselves resulting in patient centricity instead of being owned by trustless centralized servers like hospital databases. When patients are the owner of their data, they will grant permission to anyone who wants to access that data and how their data will be used [21, 22, 63, 64].

5.4 Requirement 4: traceability and data integrity

Data traceability and data integrity are critical concerns in data sharing among stakeholders. The data owner, i.e. the patient, should be able to track how their healthcare information is being shared and give their dynamic consent as per the purpose of data utilization to achieve transparency. During the process if any malicious node tries to manipulate the health record, then it should be detected and prevented from any change ensuring data integrity [23, 65–67].

5.5 Requirement 5: interoperability and standardization

Along with the implementation of the proposed model, it should be feasible to integrate it with the existing healthcare systems consisting of Electronic Health Records (EHRs) to achieve the requirement of interoperability which allows the different decentralized servers operating in different locations to interact and share information over a blockchain network. In order to achieve interoperability among various entities, there is a requirement of a standard format to capture the diabetic data and thereby ensure efficient data sharing [15, 68–71].

6 System architecture

This section discusses the detailed system architecture for blockchain-based patient-centric diabetic data sharing.

6.1 System components

The constituting components of the proposed architecture are depicted in Fig. 4.

- Users: In our architecture, we have three types of users who can interact with the system.
 - Data Owner: An individual such as a patient who is the owner of the data and grants or revokes consent to access that data as per his access policy.
 - Data Controller: An organizational entity such as the hospital or laboratory which captures the patient data by storing it in their secure off-chain databases.
 - Data Consumer: An individual or organizational entity like healthcare organizations, researchers, health insurance companies, etc. that needs patient data for their specific purposes and requests access to it.
- On-chain Components
 - Blockchain: The decentralized ledger that enables secure, immutable and transparent storing and sharing of healthcare data.

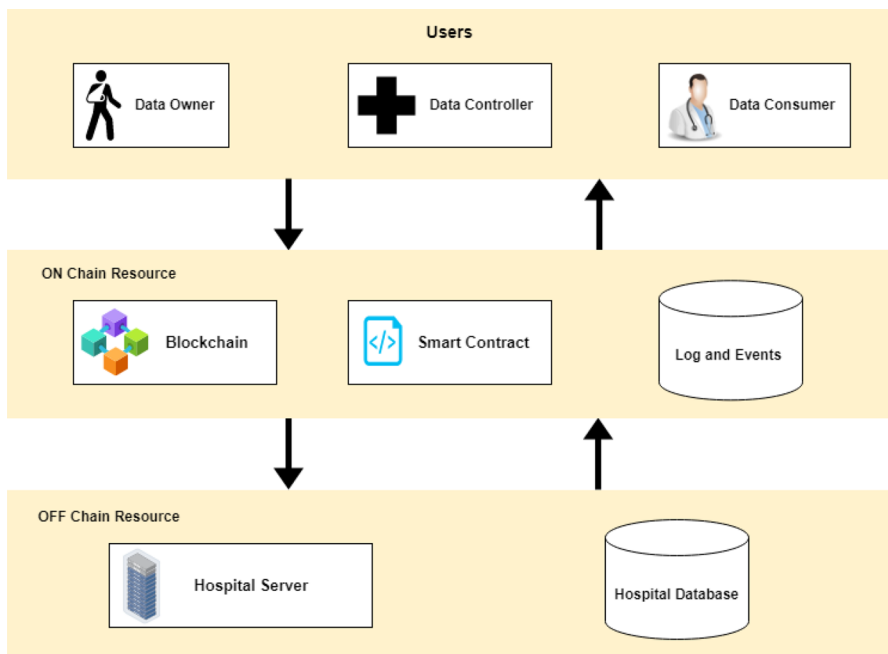


Fig. 4 System components

- Smart Contracts: The self-executing contracts which provide functionality to the system like registration of users, storing the metadata of EHR, managing access policy of patients, etc., and are written in lines of code to specify the rules of an agreement.
- Logs and Events: Functionality created by smart contracts to maintain and track the information for auditing the system data.
- Off-chain Components
 - Hospital Server: The server is responsible for managing the off-chain hospital database with the on-chain system requests.
 - Hospital Database: An off-chain database maintained for the secure storage of patient data which is managed and controlled by the data controller.

6.2 Smart contracts

In contrast to traditional contracts, smart contracts have numerous benefits such as increased automation, improved transparency, security, reduced costs and the elimination of any intermediaries. Being written in programming languages like Solidity, implementation, development and deployment of smart contracts requires certain level of technical expertise. A smart contract cannot be changed once it is deployed on the blockchain. Today, smart contracts are not considered to be legally binding and legally enforceable as their architecture still requires a legal framework component [72]. In this paper, to manage the on-chain transactions, four smart contracts have been implemented i.e. User Smart Contract, Access Policy Smart Contract, metadata smart contract and access management smart contract.

User Smart Contract (USC) USC is responsible for registration of all the users i.e. data owner, data controller and data consumer. After verification, every user has to register in USC in order to interact with the system. The Data Consumer is required to provide its public key during registration.

Access Policy Smart Contract (APSC) APSC is responsible for storing the consent policy of Data Owner. The patient (data owner) stores the set of rules in this smart contract which defines who can access their data ensuring privacy of the data.

Metadata Smart Contract (MDSC) MDSC is responsible for storing the patients' metadata that does not reveal any confidential information as a reference to the patient data stored on the secure off-chain databases.

Access Management Smart Contract (AMSC) AMSC is responsible for managing the access policy of the Data Owner by accepting access requests from Data Consumer, validating the request, notifying the Data Controller to generate one-time access URL and communicating it back to Data Consumer.

7 Proposed work

The proof-of-concept of the proposed blockchain-based architecture was implemented on a privately permissioned blockchain to showcase its viability. At the infrastructure level, Go-Ethereum [73], an open-source Ethereum client enabling

permissioned private blockchain networks, was utilized to establish a private blockchain. The system's smart contracts are written using Solidity, and the truffle framework was employed for developing, compiling, testing and deploying these smart contracts. The patient's smart contract code is partially illustrated in Fig. 5. ChainLink was used as an oracle service to connect smart contracts to off-chain resources, and MongoDB was employed to develop the off-chain database.

There are four smart contracts to handle the blockchain transactions: User Smart Contract (USC), Access Policy Smart Contract (APSC), Metadata Smart Contract (MDSC) and Access Management Smart Contract (AMSC). These smart contracts offer 6 primary system functions: createUser, setPatientConsent, cancelPatientConsent, checkPatientConsent, createMetadata and requestAccess. Only authorized users

```
pragma solidity ^0.8.0;

import "../MDSC.sol";
import "../USC.sol";
contract APSC{
    mapping(uint256 => AccessTicket) public accessTicket;

    mapping(bytes32 => bool) public accessTicketSigns;

    uint256[] public accessTicketIds;

    uint256 public lastAccessTicketId =200;

    mapping(bytes32 => Consent) private consent;
    bytes32[] public consentSigns;

    struct Consent{
        bytes32 consentSign;
        bool status;
        bytes32 MetaData;
        bytes32 role;
        bytes32 aim;
        uint256 timestamp;
        uint256[] issuedAccessTicket;
    }

    event consentlog(
        address indexed patientConsent,
        bytes32 _consentSign,
        uint256 _timestamp,
        bool indexed _status
    );
    struct AccessTicket{
```

Fig. 5 Partial code of the Access Policy Smart Contract (APSC)'s smart contract

are limited to calling these functions which are implemented using smart contract modifiers. If a function call is unauthorized, then it will result in the termination of the function execution and the reversion of all modifications to their initial state. The subsequent section elaborates on the use of smart contract functions to implement the main system functionalities.

7.1 User Smart Contract (USC)

Every system participant engages with the system using their won wallet, which contains all necessary information for system interaction. As a result, participants must be registered within the system, where a smart contract is established. The role of the system admin is to verify the users' identities and professional registrations before initiating the system registration process. Algorithm 1 describes the user registration process.

Algorithm 1 Pseudo-code for registering a new user (RegisterUser)

```

1: Input: initiator, userWalletAddress, role
2: Output: userIdentifier
3: if initiator = owner and userWalletAddress ≠ null then
4:   Require ← newUser
5:   Display "User already approved!"
6:   Create newUserProfile
7:   return userIdentifier
8: else
9:   Display an error message and undo smart contract state
10: end if

```

7.2 Access Policy Smart Contract (APSC)

This smart contract keeps patient consent as access policy tree. A consent signature is produced by hashing the access policy tree and stored in APSC. Hashing and storing patient consent in APSC allows for efficient consent status retrieval and validation. Algorithm 2 describes the process of setting the consent of the patient.

Algorithm 2 Pseudo-code for saving consent of the patient (SavePatientConsent)

```

1: Input: initiator, aim, role
2: Output: status
3: PatientConsent, consentSign ← mapping
4: if initiator = owner AND aim ≠ null AND role ≠ null then
5:   h ← hash(role, aim)
6:   if PatientConsent.contains(h) = true then
7:     Display an error message and undo smart contract state
8:   else
9:     PatientConsent.add(h, true)
10:    return true
11:  end if
12: else
13:   Display an error message and undo smart contract state
14: end if

```

7.3 Metadata smart contract (MDSC)

This smart contract contains patient metadata. Once the patient data has been collected and securely stored in an off-chain secure database such as a diabetic laboratory database, the data controller then submits the patient's metadata to the system. This metadata is a representation of the patient data that contains a description without disclosing any confidential or identifiable patient information, like the data type, hash of the stored data and conditions. The patient metadata is then kept in a separate data structure, where the hash of the stored data serves as the key and the remaining patient data as the value. Algorithm 3 shows the process of creating patient metadata.

Algorithm 3 Pseudo-code for generating patient metadata (GenerateMetadata)

```

1: Input: initiator, patientWalletAddress, dataHash
2: Output: id
3: Metadata  $\leftarrow$  mapping
4:  $i \leftarrow$  counter
5: if initiator = dataControllerWalletAddress AND patientWalletAddress  $\neq$  null
   AND dataHash  $\neq$  null then
6:    $i \leftarrow i + 1$ 
7:   Metadata.add( $i$ , [patientWalletAddress, dataHash, dataControllerWalletAd-
   dress])
8:   return  $i$ 
9: else
10:  Display an error message and undo smart contract state
11: end if

```

7.4 Access management smart contract (AMSC)

This smart contract is responsible for managing access to patient data. The Data Consumer should obtain an Access Ticket (AT) to be able to access the patient information. Using the AT, the access to the requested data is limited to the lowest possible level. Once the data consumer identifies relevant patient metadata, they have to request an AT from the system by providing the hash of the data they requested, the reason for requesting access to patient data and their role. This request is then verified by APSC, which stored the consent policy of the patient. If the consent is valid and aligns with the data consumer's request, an AT is automatically generated for data consumer. Algorithm 4 shows the process of requesting access to off-chain relevant patient data.

Algorithm 4 Pseudo-code for requesting permission for accessing off-chain patient information. GenerateMetadata (RequestAccess)

```

1: Input: dataConsumer, Metadata, aim, role
2: Output: ticketId
3: METADATA  $\leftarrow$  mapping
4: if dataConsumer = owner AND Metadata  $\neq$  null AND aim  $\neq$  null AND role
    $\neq$  null then
5:   d  $\leftarrow$  data.get(MetadataID)
6:   patient  $\leftarrow$  d.patientWalletAddress
7:   h  $\leftarrow$  hash(role, aim)
8:   if PatientConsent.get(h) == TRUE then
9:     ticket  $\leftarrow$  patient.CreateAccessTicket(dataConsumer, MetadataID)
10:    ticket.status  $\leftarrow$  TRUE
11:    return ticket.id
12:   else
13:     Display an error message and undo smart contract state
14:   end if
15: else
16:   Display an error message and undo smart contract state
17: end if

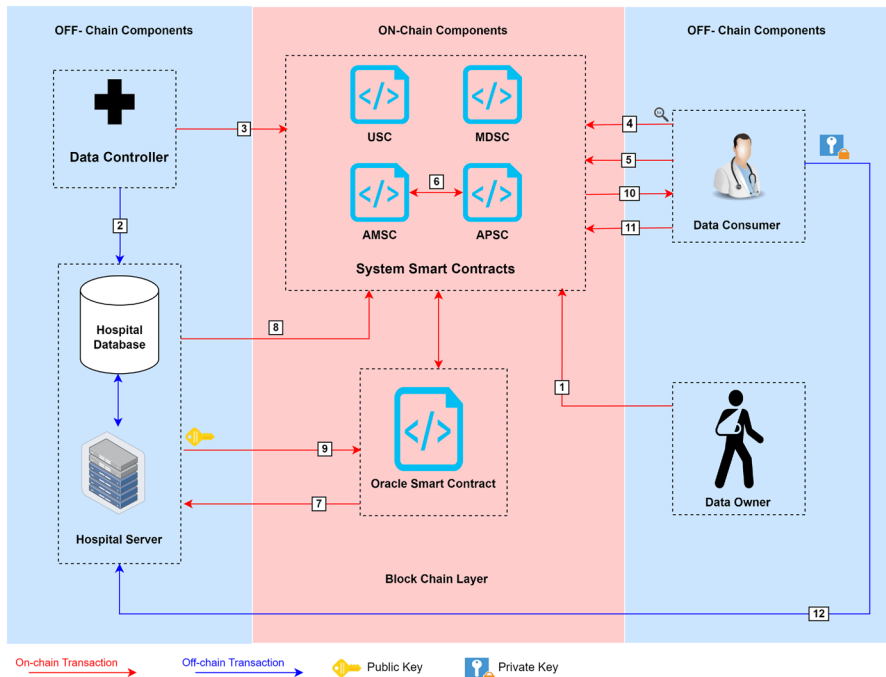
```

7.5 Proof-of-concept: DiabeticChain

In this section we present DiabeticChain, an implementation of the proposed architecture as a proof-of-concept study. DiabeticChain enables the patients to grant or revoke consent to data consumers so that they can access their diabetic data and data controllers to collect and store their diabetic data. Figure 6 depicts a high-level detailed work flow of the implementation of the proposed architecture.

All users—Data Owner, Data Controller, Data Consumer—should first register in User Smart Contract (USC). During the Data Consumer Registration, the Data Consumer should provide his public key.

1. Data Owner (patient) stores his access policy in APSC (Access Policy Smart Contract) on the blockchain.
2. Data Controller then collects the patient's diabetic data, including blood glucose levels, insulin dosages, and other relevant information and stores it in a secure off-chain hospital database.
3. Data Controller then stores patient's metadata in MDSC on the blockchain.
4. Data Consumer can search for patient data of interest from the available data in MDSC.
5. After identifying patient data of interest, Data Consumer sends an access request to AMSC.
6. AMSC validates the access request with APSC by checking the patient access policy (if there is a valid access policy move to next step 7, otherwise access request will be denied and the data cannot be accessed by the data consumer).
7. On acceptance, AMSC notifies Hospital to prepare the data for sharing with a Data Consumer, the Hospital retrieves Data Consumer's public key from USC and Data Consumer's data from the database.



All users should register in USC. During Data Consumer registration, Data Consumer should provide his public key.

1. Data owner stores his access policy in APSC on the blockchain.

2. Data Controller collects the patient's diabetic data, including blood glucose levels, insulin dosages, and other relevant information, and stores it in a secure off-chain database

3. Data Controller then stores patient metadata in MDSC on the blockchain.

4. Data Consumer can search for patient data of interest in MDSC

5. After identifying patient data of interest, Data Consumer sends an access request to AMSC.

6. AMSC validates the access request with APSC by checking the patient access policy (if there is a valid access policy move to step 7 otherwise access request will be denied)

7. AMSC notifies Hospital to prepare data for sharing with Data Consumer, Hospital retrieves Data Consumer's public key from USC and Data Consumer's data from the database.

8. Hospital retrieve data consumer's public key from USC

9. Hospital creates one-time access URL for patient data and encrypts the one-time access URL using Data Consumer's public key and then stores the encrypted one-time access in AMSC

10. AMSC notifies Data Consumer about the encrypted one-time access URL

11. Data Consumer retrieves the encrypted one-time access URL from AMSC

12. Data Consumer decrypt the encrypted one-time access URL using his corresponding private key and then it to access patient data.

Fig. 6 Detailed workflow of the proposed architecture

8. Hospital then retrieves the data consumer's public key from USC on the blockchain.
9. Hospital creates a one-time access URL for patient data and encrypts the one-time access URL using Data Consumer's public key and then stores the encrypted one-time access in AMSC.
10. AMSC notifies Data Consumer about the encrypted one-time access URL.
11. Data Consumer retrieves the encrypted one-time access URL from AMSC.

12. Data Consumer decrypts the encrypted one-time access URL using his corresponding private key and then it uses it to access patient data.

8 Results and evaluation

DiabeticChain, a blockchain-based patient-centric healthcare architecture to support diabetic data sharing, is proposed and implemented in this paper. In this section, the key findings and outcomes derived from the implementation and evaluation of the proof-of-concept DiabeticChain system are discussed by addressing the design requirements, performing the security analysis and the performance evaluation. Our results do not just showcase the functionality of DiabeticChain. Through real-world scenarios and case studies, we demonstrate its transformative potential in diabetic data management, a novelty in the current landscape.

8.1 Addressing design requirements

In this section, we discuss how our system fulfils each design requirement and addresses them by implementing various features using Ethereum blockchain and smart contracts.

8.1.1 Data security

In this paper, we have proposed that the sensitive diabetic health data should be stored in the secure off-chain database and only the metadata with a reference pointer to the source data, encrypted by public key cryptography should be stored on the blockchain. Data consumer can request access to the patient data and patient can then grant or revoke access as per their consent policy. Cryptography methods and access control mechanisms ensure the security of data in DiabeticChain, thereby preventing the risk of cyberattacks.

8.1.2 Patient privacy

Our system, DiabeticChain, preserves the patient's privacy by storing the sensitive data in secure off-chain private database, and its metadata is stored on the blockchain encrypted using cryptographic techniques like public key cryptography. Every user, i.e. data owner, data consumer and data controller, is required to register in the User Smart Contract on the blockchain to authorize and validate its participation in the network. An access control mechanism has been used to safeguard the data from leakage. When the data controller receives a validated access request from the data consumer, it encrypts the secure one-time access URL using the public key of data consumer and can be decrypted only by the respective authorized public key, thereby protecting the patient privacy. If the data consumer wants to access the same data in future, then a subsequent one-time access URL will again be generated. The anonymity of data is maintained while sharing it with various stakeholders. Patient data

is anonymized by generating random unique accounts linked to the patients using their public key. One patient can be linked to multiple accounts. These accounts are then used to send records on the blockchain maintaining the anonymity of patients, and their real identity is not disclosed. In this way, data anonymization, encryption and access control mechanisms are used for patient privacy preservation.

Any security flaws in the implementation of smart contracts can result in a potential privacy risk to the system and can lead to identity theft or security theft. Also due to the public nature of blockchain, a series of transactions can be traced and lead to revelation of original identities of the data owners [74], but this is mitigated by creation of multiple accounts for a patient. These privacy risks can be mitigated by countermeasures like eliminating any potential flaws within the system. Some of the known privacy preserving techniques like zero knowledge proofs can also be used to enhance the security of the system. Using this mechanism, information is validated using cryptographic algorithms ensuring its correctness without revealing or leaking any sensitive information [75].

8.1.3 User control over data

The proposed system DiabeticChain, following a patient-centric approach, provides complete user control over their data. Being a decentralized peer-to-peer network, there is no central authority to take control of the patient data. It is owned, managed and controlled by the patients and shared among stakeholders only with their consent. The system also allows the patients to dynamically update their consent policy as per the changing environments giving them full control over their diabetic data.

8.1.4 Traceability and data integrity

Blockchain is an immutable ledger where every node contains a copy of the stored data. Whenever a new record is added, a consensus protocol is used to verify the record before appending it to the shared ledger and a cryptographic hash is computed for every block. If any malicious node tries to change the data, it will be detected by the other nodes because even a tiniest change in the data will change the hash value of every subsequent block. Hence, it cannot be manipulated or tampered ensuring data integrity. Each transaction on the blockchain is recorded and the data owner is able to track their data and how it is being used, guaranteeing the traceability of data.

8.1.5 Interoperability and standardization

While using blockchain features, DiabeticChain models a feasible design that can be integrated with the existing healthcare systems and thus ensures interoperability among systems. Since the credibility of the new records added to the blockchain depends on the credibility of its participating nodes, a form of data standardization is followed so that all the nodes can understand the data [76]. One such standard is HL7 which is a human-readable standard to address the requirement of interoperability and ensure seamless integration and data sharing among different healthcare

entities [77]. A potential limitation of using such standards can be the mismatch of versions used among different healthcare centres or possible technological issues which need to be taken care of.

8.2 Security analysis

The security of DiabeticChain is analysed in this section based on data storage in the system, sharing of data in the system and tamper-proofing.

8.2.1 Data storage

In this paper, we propose a hybrid model for the storage of data consisting of both off-chain and on-chain storage where the confidential diabetic data of the patient is stored in an off-chain secure and private database and only the metadata, hash and a reference pointer are stored on the blockchain. During every appointment, Data Controllers like hospitals and laboratories collect and record the patients' diabetic data like the blood sugar levels, blood pressure, ongoing medicines, insulin dosage and other sensitive data. In the scope of this research, we have considered that this data is then securely stored and managed on the off-chain hospital databases. Such a large dataset cannot be stored on the blockchain itself since it is not computationally and memory efficient to manage such large datasets on the blockchain. Every node in the blockchain has a replica of the data thereby increasing the required storage and a consensus protocol cannot execute such fast transactions and hence the performance will be slowed [78]. Therefore, blockchain is used to store only the metadata in the MDSC along with a reference pointer to source data in the hospital database. This division of data thus helps in optimizing the system's performance and security. The proposed hybrid storage model thus addresses the scalability and efficiency concerns associated with blockchain-based systems, such as fulfilling the storage requirements and reduction in transaction processing time.

The storage of metadata on the blockchain gives a broader view of the entire diabetic data to the Data Consumer. Since every node in the blockchain network has a local copy of the data, data consumers can easily look for the patient data of interest in the Metadata (stored on-chain) without searching exhaustively in the large off-chain datasets. After identifying patient data of interest, Data Consumer can send an access request which is validated and a one-time access URL is generated to retrieve the off-chain secure data using the reference pointer stored on-chain. Even in the case of a server failure or any other technical issue, the hybrid storage model prevents data loss by eliminating single point of failure, handles data backup as sensitive diabetic data is stored across numerous laboratories without storing any redundant data and thus ensures data availability.

Using blockchain for data storage is beneficial over traditional relational databases (RDBs) in building trust, security and privacy aspects. But compared to blockchain, RDBs offer a higher throughput in writing data [79]. Hence, a hybrid model is the most suitable option for data storage in DiabeticChain healthcare system that leverages the benefits of both blockchain and relational databases.

8.2.2 Data sharing

Patient data is stored on the off-chain database and is effectively shared among healthcare entities like researchers, doctors, laboratory technicians, etc. using DiabeticChain. Only authenticated users can request to access the sensitive diabetic data and can only access that data using a one-time access URL within a given timeframe according to the self-executing conditions coded in the smart contracts. DiabeticChain system also allows the patients to dynamically update their consent allowing the patient to grant or revoke real-time access to their data at any point. The data access request of the data consumer is verified only as per the consent policy of the patient stored in Access Policy Smart Contract (APSC).

When a hospital receives a validated request from data consumer, the hospital generates a JSON format file that consists of the requested patient data and this file is then stored on the off-chain database at a temporary location. It can be accessed using HTTPS but only once within a given short timeframe. Now, the hospital generates a one-time access URL and encrypts it using the data controller's public key obtained from the User Smart Contract (USC). This URL is then stored on the blockchain and upon decryption can direct the user to the JSON file containing the required data. The URL can be only decrypted using the corresponding private key and used to access the data once. So, the Data consumers obtain the URL from the blockchain, decrypt it using its private key and access the data. Once the URL is used to access the data, it becomes useless as the JSON file is removed from the temporary storage and now that URL will lead to nothing. Even if someone retrieves the private key of data consumer to decrypt the URL, the patient data will still be safe. If the one-time access URL is not used within the given timeframe, then also the JSON file will be removed from the temporary storage and the URL expires to prevent unauthorized access attempts. If a need to access the same data arises in the future, data consumer must again submit the request to access the data and a new URL will be generated. This process allows the secure sharing of diabetic data with the patient's consent.

The DiabeticChain system maintains a record of data sharing transactions like who accessed the diabetic data, time of accessing the data and its purpose, and this traceability of the stored diabetic data contributes to the overall security and transparency of the data sharing process by ensuring patient's trust in healthcare providers. In case of multiple data consumers leading to large number of data sharing requests, high volumes of data are shared and the system may face an issue of scalability, but DiabeticChain provides a reasonable performance, efficiency and scalability with the current version of Ethereum blockchain implemented and hybrid model of data storage used.

8.2.3 Tamper-proofing

Mechanisms and features that make the DiabeticChain system tamper-proof are the immutability of blockchain, the use of consensus algorithms to add new records and cryptographic hashing. Blockchain is an immutable ledger that cannot be changed and generation of a new block requires consensus among all the participating nodes

to prevent tampering with data. It is difficult for any malicious node to change the patient records owing to the cryptographic hash used with every block, and it forbids any manipulation or modification of the data. But, cyberattacks like Sybil attacks, man-in-the-middle attacks, denial of service attacks and 51% attacks may pose a potential risk to the system.

Sybil Attack is where identity theft occurs in a peer-to-peer network and some node is completely cut-off from their neighbouring nodes, thereby establishing connections with the attacker and this node is then fed malicious information [80]. As a countermeasure, our blockchain system can be made attack resistant by guaranteeing at least a single honest surrounding node. Another potential attack is the man-in-the-middle attack in which a malicious user may tamper with the information communicated between two nodes. Increasing the number of validations before verifying a transaction, having multiple paths available on the network and selecting many nodes to check the transaction status can act as countermeasures [81]. Denial of service attack is another threat where authorized requests are prevented by sending requests more than the handling capacity of victim server. As a countermeasure, clients are required to solve puzzles before getting the access to a resource or server [82]. In the 51% attack, the attacker gains access to 51% of the network's computational power while using proof-of-work consensus protocol or it gains control over 51% of the participating nodes when proof-of-authority is used. To prevent these attacks, a proof-of-authority consensus algorithm is used where the validators stake their reputation and it is much more difficult for the attacker to control 51% of the nodes in the network, thus minimizing the risk of potential cyberattacks.

8.3 Performance evaluation

To assess the performance of DiabeticChain, we utilized 4 virtual machines (VMs) hosted on Microsoft Azure cloud to establish a private blockchain using Go-Ethereum, an open-source Ethereum client enabling permissioned private blockchain networks. Hyperledger Caliper, an open-source benchmarking tool for evaluating blockchain applications' performance, was employed to conduct multiple tests, validating DiabeticChain's effectiveness. Table 5 displays the private blockchain and testing environment configurations. We tested two primary types of blockchain operations, Transaction and Query operations, using four performance metrics to evaluate DiabeticChain: Transaction throughput, Query throughput, Transaction latency and Query latency [83].

We conducted 10 test rounds and varied the transactions per second (tps) gradually from 100 to 1000 with an increase of 100 tps with every round, and hence, the send rate increased linearly from 99.38 tps to 748.22 tps for query operations and from 100.1 tps to 839.76 tps for transaction operations.

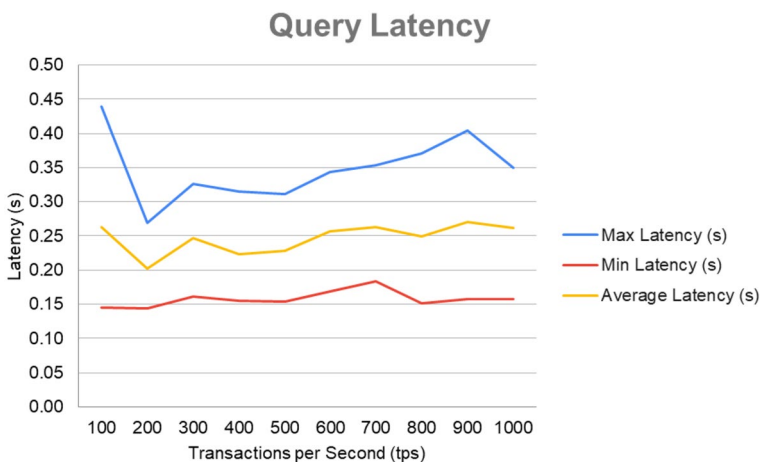
Figures 7 and 8 demonstrate the Query Latency and Throughput Latency, respectively. Each chart consists of the Maximum Latency, Minimum Latency and the Average Latency of the corresponding operations. DiabeticChain was able to run the Query operations efficiently with low latency, and Transaction operations

Table 5 Go-Ethereum and testing environment configurations

Factor	Setting
Nodes	4 VMs with Ubuntu 20.04 on Microsoft Azure cloud, every VM equipped with 2 Intel CPUs and 8 GB RAM.
Blockchain	3 peer nodes 1 validator node Go-Ethereum v1.10.11
Consensus	PoA (Clique Protocol)
Smart contracts	Solidity
Benchmarking Tool	Hyperledger Caliper v0.5.0

were handled at a higher latency as they involve more complex operations of changing the data on the blockchain. The average Query Latency was 0.246 s (Fig. 7) and the average Transaction Latency was 1.31 s (Fig. 8). The maximum and minimum Query Latency was obtained to be 0.35 s and 0.16 s, respectively. For the Transaction operations, the maximum and minimum latency was 1.75 s and 0.88 s, respectively.

Figures 9 and 10 demonstrate the Query Throughput and Transaction Throughput, respectively. An average query throughput of 201.7 tps was obtained by varying the send rate linearly from 99.38 to 748.22 tps within 10 rounds. The maximum query throughput of 243.54 tps was achieved when the send rate was 748.22 tps, while the minimum query throughput was 78.8 tps when the send rate was 99.38 tps. Similarly, the send rate was varied from 100.1 to 839.76 tps in 10 rounds of testing to achieve an average transaction throughput of 200.07 tps. The maximum and minimum average transaction throughput obtained was 246 tps and 80.54 tps at the send rates of 673.94 tps and 100.1 tps, respectively.

**Fig. 7** Query latency

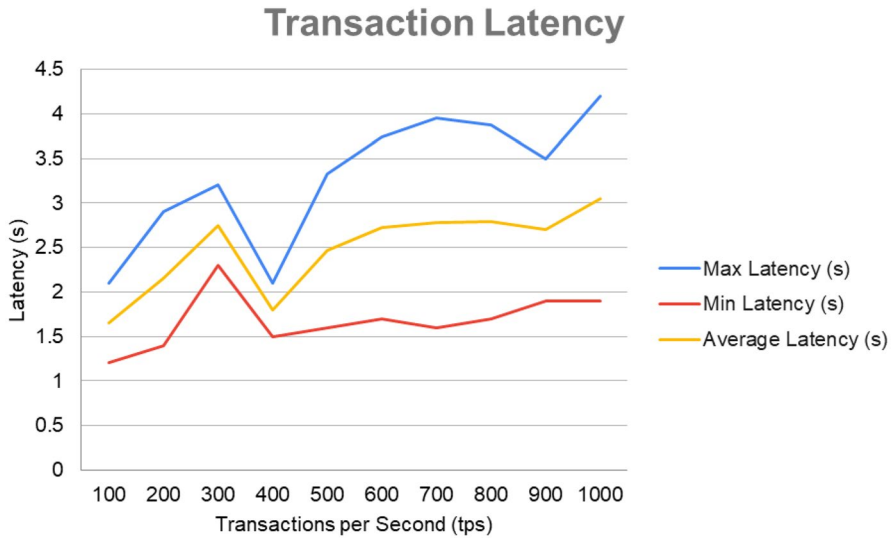


Fig. 8 Transaction latency

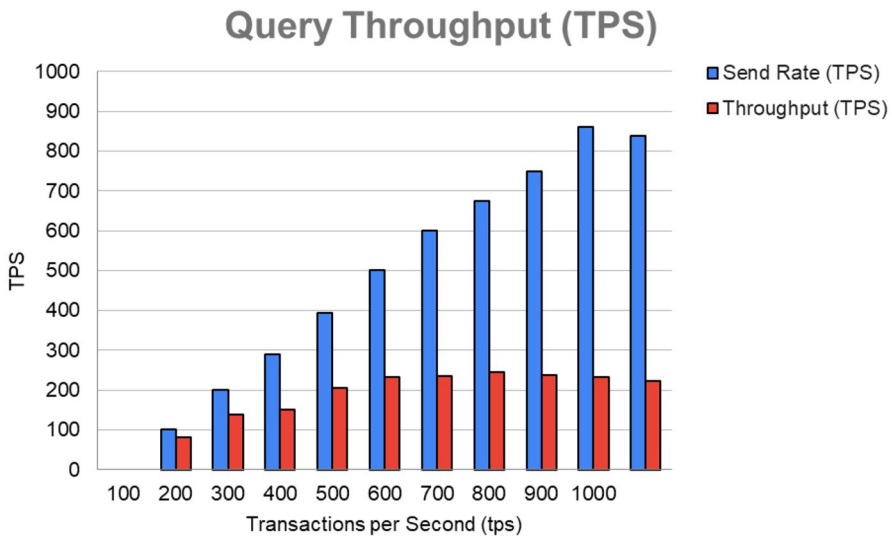


Fig. 9 Query throughput

9 Discussions and conclusion

This section discusses the advantages of the proposed architecture, the current limitations of DiabeticChain and the future work which can further enhance the implementations of the proposed system, followed by a conclusion.

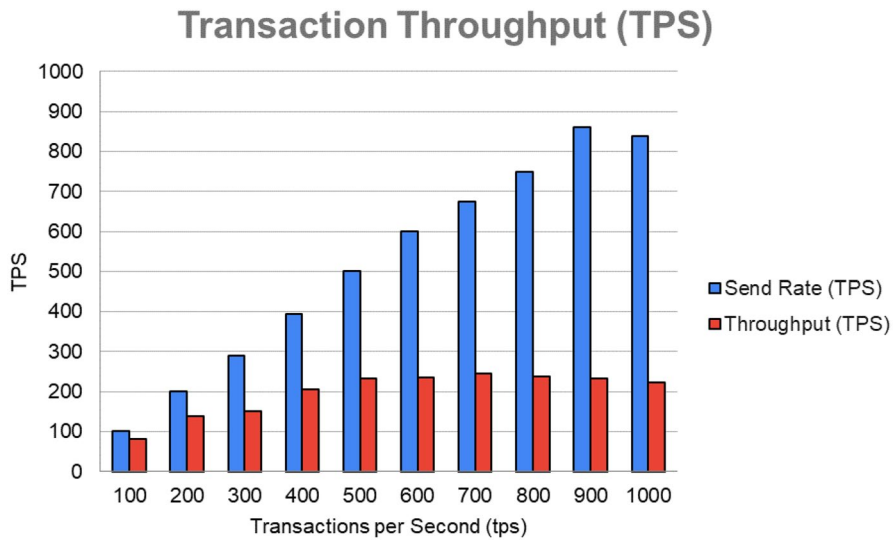


Fig. 10 Transaction throughput

9.1 Advantages of the proposed architecture

DiabeticChain is a patient-centric healthcare architecture based on blockchain technology to support secure and efficient diabetic data sharing. The proposed architecture is implemented to overcome the current challenges of traditional data sharing methods like lack of privacy, security, data ownership by patients, interoperability, fragmentation of data and data integrity. The key advantages offered by the implementation of the proposed system as a proof-of-concept study to demonstrate DiabeticChain's effectiveness are as follows:

- Unlike traditional systems that compromise on data security, DiabeticChain stands out. Our hybrid model ensures not just secure storage but also optimizes data retrieval, a first in the domain.
- Secondly, the system minimizes the risk of security breaches with the use of cryptography techniques like public key cryptography and access control mechanisms involving the patients access policy to grant or revoke access to their data.
- Thirdly, the system preserves the privacy of the patient by using a one-time access URL to access the diabetic data while maintaining the anonymity of data by creating random unique accounts associated with the patients.
- The system allows the patients to dynamically update their consent to grant or revoke data consumer's access, thereby giving them complete control over their diabetic data due to the absence of any centralized authority.

- While leveraging the benefits of blockchain technology, the system additionally ensures data integrity using proof-of-authority consensus protocol, immutability of blockchain and thereby enables traceability of data.
- Finally, the system provides a feasible model which can be integrated with the existing healthcare systems and thus ensures interoperability.

9.2 Limitations and future work

The current implementation of DiabeticChain uses a privately permissioned blockchain, and it may pose a potential risk as private blockchains have a lesser number of nodes in the network. So, it becomes easier for a malicious node to acquire control over the network and the risk of data breaches and manipulation of data is higher as compared to public blockchains [84]. The proposed system uses a one-time access URL for data sharing which should be accessed within a timeframe and can pose another limitation to the current implementation. The system can also pose legal and regulatory issues as there does not exist any central authority that can be held accountable for the blockchain network. There exist potential barriers to the adoption of DiabeticChain due to healthcare professionals' acceptance of the technology and technical barriers like the requirement of a stable internet connection, the establishment of every healthcare entity on the network and the requirement of the necessary hardware.

In the future, the current implementation of the proposed system can further be integrated with other emerging technologies like Internet of Things (IoT) to directly capture continuous patient data from smart wearable devices like smart watches to have daily smart care and artificial intelligence (AI) to make use of intelligent agents to interact with the system and further ease the process of data sharing. A potential enhancement to the current architecture can be to integrate machine learning algorithms with the captured data for predictive analysis of diabetic data like early detection of diabetes and enhanced cure. Another possible use of ML and AI can be to analyse the abnormal node behaviours within the network. There is a need to assess the performance and scalability of the system with an increasing number of users working simultaneously on the blockchain. Certain technological barriers may exist which hinder the process to completely shift the traditional Electronic Health Records (EHR) data to blockchain-based systems and fully implement blockchain-based solutions, but this to happen in reality does not seem very far.

9.3 Conclusions

Sensitive diabetic data is generated in large amounts and needs to be shared effectively among stakeholders since diabetes requires significant self-care and substantially burdens the patients with processes like continuous glucose monitoring and carbohydrate counting. The controlled access and sharing of diabetic data play a crucial role in benefitting healthcare professionals to take better patient care and researchers to effectively study disease patterns. In this paper, we addressed the current challenges associated with the privacy, security and controlled sharing of sensitive diabetic data, thereby improving patient care and advancing research. Our research identifies these

gaps, laying the foundation for the novel solutions DiabeticChain offers. We recognized the current limitations of the traditional client–server and cloud-based approaches such as privacy, security concerns, data fragmentation, authorization, lack of interoperability and data integrity and hence proposed a private permissioned blockchain-based solution for sharing of diabetic data.

DiabeticChain, our blockchain-based solution to diabetic data sharing, provides a novel patient-centric approach, leveraging access control mechanisms and smart contracts written in solidity to facilitate secure and efficient sharing of diabetic data among stakeholders while addressing concerns regarding privacy, security, data fragmentation, authorization and data integrity. We implemented a proof of concept of the architecture proposed using a privately permissioned Ethereum blockchain and analysed the system's performance where the results depicted an average Query throughput of 201.7 tps and an average Transaction throughput was 200.07 tps. The average Query latency was 0.24 s and the average Transaction latency was 1.31 s proving DiabeticChain to be scalable and efficient.

In comparison with the existing blockchain-based healthcare systems, we propose a blockchain-based architecture that uses a one-time access URL generated as per the access policy of the data owner, improving privacy, security and efficiency of diabetic data sharing. The proposed architecture will have a potential impact on patient care and research in the context of diabetes management as well as integration and application in other healthcare domains by enhancing the efficiency of the data sharing process, ensuring patient control over their data, and enabling privacy and data security. Through real-world scenarios and case studies, we demonstrate DiabeticChain's transformative potential in diabetic data management, a novelty in the current landscape.

Author contributions Deepak Kumar Mishra helped in conceptualization, formal analysis, writing original draft. Pawan Singh Mehra was involved in supervision, methodology, validation.

Data availability Not applicable.

Declarations

Conflict of interest None.

Ethical approval Yes.

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